

Théo Morales, Ph.D. in Computer Science

3D Computer Vision & Machine Learning

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🌐 Théo Morales

🌐 <http://theomorales.com/>



Postdoc researcher with a background in Computer Science & Engineering and a **focus on 3D/4D reconstruction and Gaussian Splatting**. Bringing over six years of experience in computer vision, complemented by hands-on expertise in model training and deployment. Successfully contributed to cutting-edge research at Trinity College Dublin, **showcasing a deep understanding of diffusion models, latent variable models and 3D/4D representations**. Eager to leverage strong analytical skills and innovative thinking to drive future advancements in technology.

Employment history

- 2024 – 2026 ■ **Postdoctoral researcher.** School of Computer Science & Statistics, Trinity College Dublin, Ireland.
Research: 4D reconstruction; dynamic Gaussian Splatting; 4D representations
- 2023 ■ **Short-term lecturer.** Fuzhou University & Maynooth University, China.
- 2020 – ... ■ **Programming Tutor.** online and in Dublin, Ireland.
- 2019 – 2020 ■ **Computer Vision engineer (R&D).** Dopli / Cisteme / XLIM, France.
Topics: edge AI; object tracking; model training & deployment
- 2019 ■ **Research assistant.** Artificial Intelligence in Robotics group, Aarhus University, Denmark.
Topics: ROS; Gazebo; sim2real; UAV robotics
- 2017 – 2019 ■ **Part-time software developer.** Trifork, Aarhus, Denmark.
Topics: web development; iOS development
- 2016 – 2017 ■ **Part-time software developer.** Asimut Software, Aarhus, Denmark.
Topics: full-stack development

Education

- 2020 – 2024 ■ **Ph.D. Computer Science, Trinity College Dublin.**
Thesis title: *Designing Hand-Object Interaction Representations for Better Grasp Priors*.
Topics: meta-learning; generative models; diffusion models; 3D representations
- 2017 – 2019 ■ **M.Sc. Computer Engineering, Aarhus University.**
Thesis title: *Synthetic Images for Convolutional Neural Networks in Autonomous Drone Racing*.
Topics: CNNs; ROS; sim2real; OpenGL; robotics
- 2016 – 2017 ■ **B.Eng. Information and Communication Technology, VIA University College.**
Thesis title: *Pizza Ordering Management System*.
Topics: TDD; REST API
- 2013 – 2016 ■ **DUT. Information and Communication Technology, IUT La Rochelle / VIA University College.**
Undergraduate IT diploma in Embedded Systems & Signal Processing.

Publications

Conference proceedings

- 1 T. Morales, O. Taheri, and G. Lacey, “A versatile and differentiable hand-object interaction representation,” in *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2025, pp. 23–33. [DOI: 10.1109/WACV61041.2025.00013](#).
- 2 S. Ravi, P. Climent-Perez, T. Morales, C. Huesca-Spaurani, K. Hashemifard, and F. Florez-Revuelta, “ODIN: An OmniDirectional INdoor dataset capturing Activities of Daily Living from multiple synchronized modalities,” in *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, Los Alamitos, CA, USA: IEEE Computer Society, Jun. 2023, pp. 6488–6497. [DOI: 10.1109/CVPRW59228.2023.00690](#).
- 3 T. Morales and G. Lacey, “A new benchmark for group distribution shifts in hand grasp regression for object manipulation. can meta-learning raise the bar?” In *NeurIPS 2022 Workshop on Distribution Shifts: Connecting Methods and Applications*, 2022. [URL: https://openreview.net/forum?id=IKbA3QS7c8X](#).
- 4 T. Morales, A. Sarabakha, and E. Kayacan, “Image generation for efficient neural network training in autonomous drone racing,” in *2020 International Joint Conference on Neural Networks (IJCNN)*, 2020, pp. 1–8. [DOI: 10.1109/IJCNN48605.2020.9206943](#).

Skills

Expertise	Mathematics for ML, Computer Vision, Computer Graphics, 4D reconstruction, 3D Gaussian Splatting, Diffusion Models, Transformers, ...
MLOps	Kubeflow, Docker, Kubernetes
Languages	Native in French, bililingual in English, near working proficiency in Spanish.
Programming	C, C++, Python, Java, Zig, CUDA, Bash, ...
Misc.	Academic research, teaching, training, consultation, $\LaTeX$ typesetting and publishing.

Miscellaneous experience

Achievements

2025 [Awarded funding](#), Research funding for a project with Dolby Research.

Conference reviewing

2024 – ... [Reviewer](#). CVPR, NeurIPS.

References

Available on request.